Laboratory Equipment and Safety

Answer key with worked solutions

A. Hazards

1. Pictograms.

> (a) **Flammable**: ethanol (b) **Oxidizer**: potassium nitrate or concentrated hydrogen peroxide (c) **Corrosive**: concentrated acids or sodium hydroxide (d) **Acutely toxic**: mercury compounds (e) **Toxic to aquatic life**: copper(II) sulfate. Other correct examples are acceptable.

2. Signal words and the SDS.

> (a) **Danger** (Warning is less severe). (b) A **safety data sheet** gives a chemical's hazards, safe handling, storage, first aid and disposal. Read it **before** using the chemical, as part of the risk assessment.

3. Five unsafe actions.

> 1. Goggles pushed up: keep **goggles over the eyes** whenever chemicals are in use.
> 2. Water poured into concentrated acid: add the **acid slowly to water**.
> 3. Ethanol heated over a flame: it is flammable, so use a **water bath** or hot plate.
> 4. Test tube pointed at her partner: point it **away from everyone**.
> 5. Drinking in the lab: **no eating or drinking**. (Also: copper(II) sulfate is toxic to aquatic life, so it goes in the **waste container**, not down the sink.)

B. Choosing equipment

4. Equipment.

> (a) **25 mL volumetric pipette** (b) **measuring cylinder** (or beaker) (c) **250 mL volumetric flask** (d) **burette** (e) **analytical balance**

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5. Percent uncertainty.

> (a) $\frac{0.01\ \text{g}}{0.50\ \text{g}} \times 100\% = \mathbf{2\%}$ and $\frac{0.0001\ \text{g}}{0.5000\ \text{g}} \times 100\% = \mathbf{0.02\%}$
> (b) $\frac{0.5\ \text{mL}}{10.0\ \text{mL}} \times 100\% = \mathbf{5\%}$ and $\frac{0.02\ \text{mL}}{10.00\ \text{mL}} \times 100\% = \mathbf{0.2\%}$
> The analytical balance and the pipette are about 100 and 25 times more precise. Small quantities need precise equipment, and a large cylinder is a poor choice for a small volume.

C. Burette readings

6. Titres.

> (a) Rough: 24.10 mL − 0.00 mL = **24.10 mL**. Run 1: 24.70 mL − 1.20 mL = **23.50 mL**. Run 2: 23.85 mL − 0.40 mL = **23.45 mL**. Run 3: 23.55 mL − 0.05 mL = **23.50 mL**.
>
> (b) Mean of runs 1–3:
>
> $$\bar{V} = \frac{(23.50 + 23.45 + 23.50)\ \text{mL}}{3} = \frac{70.45\ \text{mL}}{3} = 23.483\ \text{mL}$$
>
> Deviations: +0.017 mL, −0.033 mL, +0.017 mL.
>
> $$s = \sqrt{\frac{1.67 \times 10^{-3}\ \text{mL}^2}{3 - 1}} = 0.029\ \text{mL}$$
>
> Result: **23.48 ± 0.03 mL (n = 3)**. The rough titre is left out: it only shows roughly where the end point is.

7. The meniscus.

> Water forms a curved surface that dips in the middle; the calibration marks are set to the **bottom** of the curve. With the eye above or below, the meniscus lines up with the wrong mark: this is **parallax error**.

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D. Challenge

8. Titre uncertainty.

> (a) Two readings: $2 \times 0.05\ \text{mL} = 0.10\ \text{mL}$.
>
> $$\frac{0.10\ \text{mL}}{23.50\ \text{mL}} \times 100\% = \mathbf{0.43\%}$$
>
> (b) Make the titre **larger**, for example by using a more dilute titrant or a larger volume of the solution being titrated, so the same ±0.10 mL is a smaller fraction of it.

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